system reliability. A temperature sensor might cost $20, but if it requires a $1,000 cellular modem and $10 monthly service fee to transmit its data, it might be beyond your budget. Conversely, choosing a low-cost communication protocol might limit you to sensors from specific manufacturers or require deploying your own network infrastructure. These dependencies can have wide ranging impact; selecting LoRaWAN for its long range and low power might mean accepting lower data rates, which in turn limits you to simple environmental sensors rather than vibration monitors that generate rich frequency data. The communication landscape has evolved dramatically over the past decade, moving from expensive proprietary protocols to a rich ecosystem of standards-based options. This abundance creates its own challenges of how to choose from the available range of viable technologies.

3.3.1 Understanding the communication landscape

Wireless communication technologies exist on a spectrum, trading off data rate, range, and power consumption, as shown in figure 3.4:

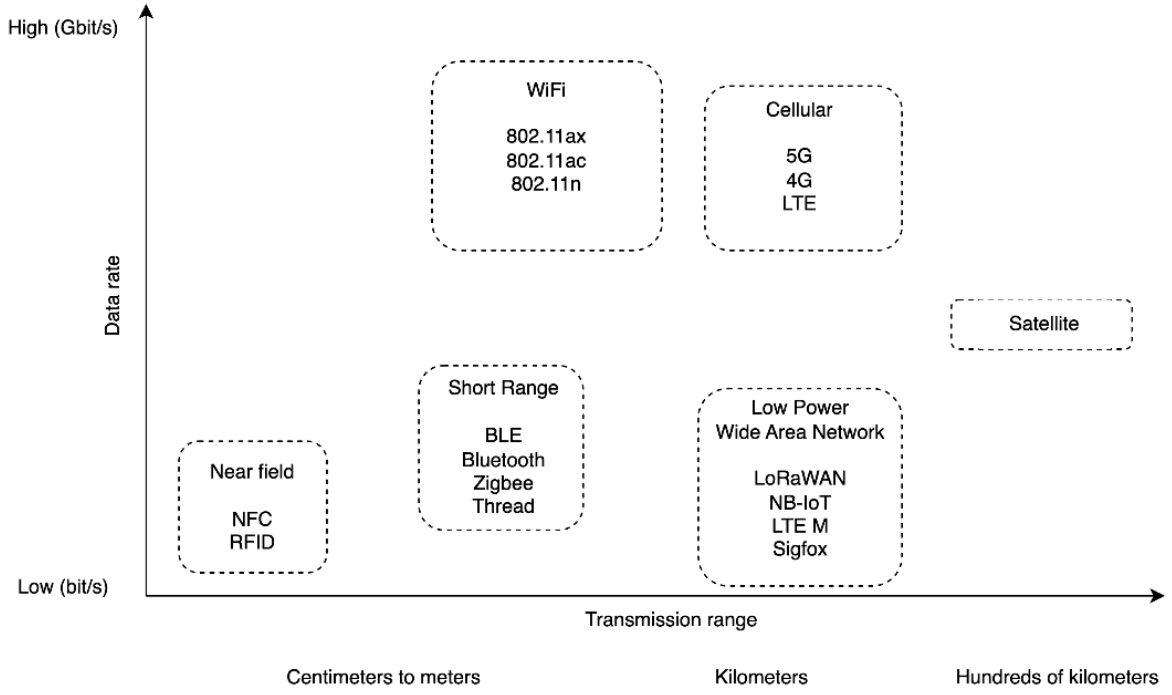


Figure 3.4 Wireless communication technology exists on a spectrum of data rate, transmission range, and cost. Higher data rates work over shorter distances and require more power, whereas longer range communication typically means accepting lower data rates.